

## SAFETY DATA SHEET

Revision Date 02-April-2024

Revision Number 4

### 1. Identification

**Product Name** Platinum, nominally 40% on durable carbon support

**Cat No. :** 47388

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                  |            |
|------------------|------------|
| Flammable solids | Category 2 |
| Carcinogenicity  | Category 2 |

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Flammable solid  
Suspected of causing cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use

Do not handle until all safety precautions have been read and understood

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Ground/bond container and receiving equipment

Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF exposed or concerned: Get medical advice/attention

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component    | CAS-No    | Weight % |
|--------------|-----------|----------|
| Carbon black | 1333-86-4 | 60.0     |
| Platinum     | 7440-06-4 | 40       |

### 4. First-aid measures

|                                        |                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                       |                                                           |
|---------------------------------------|-----------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | approved class D extinguishers. Do not use water or foam. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                  |
| <b>Flash Point</b>                    | No information available                                  |
| <b>Method -</b>                       | No information available                                  |

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Platinum oxide. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
0

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
-

**6. Accidental release measures**

|                                             |                                                                                                                                                                  |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                                                |
| <b>Environmental Precautions</b>            | Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.                                                     |

**7. Handling and storage**

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing. Avoid dust formation. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Acids. Organic materials. Oxidizing agent.                                         |

**8. Exposure controls / personal protection****Exposure Guidelines**

| Component    | Alberta                    | British Columbia         | Ontario TWAEV            | Quebec                   | ACGIH TLV                | OSHA PEL                                                           | NIOSH                                                                                    |
|--------------|----------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Carbon black | TWA: 3.5 mg/m <sup>3</sup> | TWA: 3 mg/m <sup>3</sup> | TWA: 3 mg/m <sup>3</sup> | TWA: 3 mg/m <sup>3</sup> | TWA: 3 mg/m <sup>3</sup> | (Vacated) TWA: 3.5 mg/m <sup>3</sup><br>TWA: 3.5 mg/m <sup>3</sup> | IDLH: 1750 mg/m <sup>3</sup><br>TWA: 3.5 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> |
| Platinum     | TWA: 1 mg/m <sup>3</sup>   | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup>                                 | TWA: 1 mg/m <sup>3</sup>                                                                 |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** None under normal use conditions.

**Personal protective equipment**

|                        |                                                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
| <b>Hand Protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                   |

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Natural rubber | See manufacturers recommendations | -               | Splash protection only |
| Nitrile rubber |                                   |                 |                        |
| Neoprene       |                                   |                 |                        |
| PVC            |                                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

#### Environmental exposure controls

No information available.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b>Physical State</b>                         | Solid                    |
| <b>Appearance</b>                             | No information available |
| <b>Odor</b>                                   | No information available |
| <b>Odor Threshold</b>                         | No information available |
| <b>pH</b>                                     | Not applicable           |
| <b>Melting Point/Range</b>                    | No data available        |
| <b>Boiling Point/Range</b>                    | No information available |
| <b>Flash Point</b>                            | No information available |
| <b>Evaporation Rate</b>                       | Not applicable           |
| <b>Flammability (solid,gas)</b>               | No information available |
| <b>Flammability or explosive limits</b>       |                          |
| Upper                                         | No data available        |
| Lower                                         | No data available        |
| <b>Vapor Pressure</b>                         | No information available |
| <b>Vapor Density</b>                          | Not applicable           |
| <b>Specific Gravity</b>                       | No information available |
| <b>Solubility</b>                             | No information available |
| <b>Partition coefficient; n-octanol/water</b> | No data available        |
| <b>Autoignition Temperature</b>               | No information available |
| <b>Decomposition Temperature</b>              | No information available |
| <b>Viscosity</b>                              | Not applicable           |

## 10. Stability and reactivity

|                        |                                            |
|------------------------|--------------------------------------------|
| <b>Reactive Hazard</b> | None known, based on information available |
| <b>Stability</b>       | Stable under normal conditions.            |

|                                         |                                                                         |
|-----------------------------------------|-------------------------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Incompatible products.                                                  |
| <b>Incompatible Materials</b>           | Acids, Organic materials, Oxidizing agent                               |
| <b>Hazardous Decomposition Products</b> | Platinum oxide, Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ) |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                |
| <b>Hazardous Reactions</b>              | None under normal processing.                                           |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Mist LC50** Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.

#### Component Information

| Component    | LD50 Oral                  | LD50 Dermal              | LC50 Inhalation                          |
|--------------|----------------------------|--------------------------|------------------------------------------|
| Carbon black | LD50 > 15400 mg/kg ( Rat ) | LD50 > 3 g/kg ( Rabbit ) | LC50 > 4.6 mg/m <sup>3</sup> ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component    | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------|-----------|------------|------------|------------|------------|------------|
| Carbon black | 1333-86-4 | Group 2B   | Not listed | A3         | X          | A3         |
| Platinum     | 7440-06-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

*IARC (International Agency for Research on Cancer)*

*IARC (International Agency for Research on Cancer)*

*Group 1 - Carcinogenic to Humans*

*Group 2A - Probably Carcinogenic to Humans*

*Group 2B - Possibly Carcinogenic to Humans*

*A1 - Known Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Animal Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*Mexico - Occupational Exposure Limits - Carcinogens*

*Mexico - Occupational Exposure Limits - Carcinogens*

*A1 - Confirmed Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Confirmed Animal Carcinogen*

*A4 - Not Classifiable as a Human Carcinogen*

*A5 - Not Suspected as a Human Carcinogen*

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and** No information available

delayed

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component    | Freshwater Algae | Freshwater Fish | Microtox   | Water Flea                                |
|--------------|------------------|-----------------|------------|-------------------------------------------|
| Carbon black | Not listed       | Not listed      | Not listed | EC50: > 5600 mg/L, 24h<br>(Daphnia magna) |

**Persistence and Degradability** Insoluble in water May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1325  
 Proper Shipping Name Flammable solid, organic, n.o.s.  
 Technical Name (CARBON, Platinum powder)  
 Hazard Class 4.1  
 Packing Group II

### TDG

UN-No UN1325  
 Proper Shipping Name Flammable solid, organic, n.o.s.  
 Hazard Class 4.1  
 Packing Group II

### IATA

UN-No UN1325  
 Proper Shipping Name Flammable solid, organic, n.o.s.  
 Hazard Class 4.1  
 Packing Group II

### IMDG/IMO

UN-No UN1325  
 Proper Shipping Name Flammable solid, organic, n.o.s.  
 Hazard Class 4.1  
 Packing Group II

## 15. Regulatory information

**All of the components in the product are on the following Inventory lists:** China X = listed Australia U.S.A. (TSCA) Canada (DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS)

### International Inventories

| Component    | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS    | NLP |
|--------------|-----------|-----|------|------|-----------------------------------------------|-----------|-----------|-----|
| Carbon black | 1333-86-4 | X   | -    | X    | ACTIVE                                        | 215-609-9 | 435-640-3 | -   |
| Platinum     | 7440-06-4 | X   | -    | X    | ACTIVE                                        | 231-116-1 | -         | -   |

| Component    | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Carbon black | 1333-86-4 | X     | X        | X    | X    | X    | X    | X     | X     |
| Platinum     | 7440-06-4 | X     | KE-28808 | X    | -    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component    | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA)<br>- List of Toxic Substances | Canada's Chemicals Management Plan (CEPA)         |
|--------------|------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------|
| Carbon black |                                                      |                                                                               | Subject to Monitoring and Surveillance Activities |

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Carbon black | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**<https://echa.europa.eu/substances-restricted-under-reach>**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component    | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Carbon black | 1333-86-4 | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Platinum     | 7440-06-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component    | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Carbon black | 1333-86-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Platinum     | 7440-06-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

Prepared By

Product Safety Department

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www.thermofisher.com

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Revision Date</b>    | 02-April-2024                                      |
| <b>Print Date</b>       | 02-April-2024                                      |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**